Journal of Financial Crises

Volume 7 | Issue 4

2025

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Recommended Citation
Aldridge, Patrick; Cimon, David A.; and Vala, Rishi (2025) "Central Bank Crisis Interventions: A Review of the Recent Literature on Potential Costs," *Journal of Financial Crises*: Vol. 7 : Iss. 4, 1-26.
DOI: https://doi.org/10.17132/2693-3179.1645
Available at: https://elischolar.library.yale.edu/journal-of-financial-crises/vol7/iss4/1

Cover Page Footnote
We would like to thank Jason Allen and Jon Witmer at the Bank of Canada, and Angela Vossmeyer, the editor at the Journal of Financial Crises, for their helpful comments and suggestions. The views of this paper do not represent those of the Bank of Canada. All errors are our own.

# Central Bank Crisis Interventions: A Review of the Recent Literature on Potential Costs

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**Abstract**

Central banks may engage in large-scale lending and asset purchases to stabilize financial markets and implement monetary policy during crises. The ability of these actions to restore financial market functioning is well documented; however, they come with costs. We provide a literature review of the costs associated with these post–Global Financial Crisis central bank actions, without commenting on the net benefits they provide. We find support for the premise that crisis actions may negatively impact market liquidity, distort asset prices, increase rent-seeking and inefficient uses of the liquidity provided by the central bank, create international spillovers, and create distortions in the consolidated government balance sheet. We discuss measures that may mitigate the negative impacts of crisis actions.

**Keywords:** central banks, financial crises, financial stability, lending programs, quantitative easing, repo markets

**JEL Classifications:** E5, E58, G10, G20

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[4] We would like to thank Jason Allen and Jon Witmer at the Bank of Canada and Angela Vossmeyer, articles editor at the *Journal of Financial Crises*, for their helpful comments and suggestions. The views of this paper do not represent those of the Bank of Canada. All errors are our own.

# I. Introduction

Central bank interventions in a financial crisis can be effective at both maintaining market functioning and implementing unconventional monetary policy. Lending programs by central banks can ensure solvent market participants are able to access liquidity in exchange for good collateral, one of the original recommendations from Bagehot (1873). Asset purchases can both guarantee the liquidity of relevant assets and serve as a key unconventional monetary policy tool. Bernanke (2022) argues that, in the latter case, quantitative easing (QE) can add between 3 percentage points (pp) and 4 pp to the central bank's effective lower bound for interest rates.

Liquid government bond markets are key to well-functioning financial systems. A high level of liquidity in these markets can ensure participants are confident in their ability to access liquidity in exchange for high-quality assets. In turn, there is a lower probability of asset fire sales and contagion between financial institutions. The ability to avoid these negative outcomes means that central bank interventions, which maintain the liquidity of high-quality assets, can create strong benefits for the financial systems.

Despite these benefits, large-scale actions such as asset purchases and lending facilities at central banks are bound to have side effects and unintended consequences. An analysis of central bank interventions requires a proper accounting of not only the benefits but also these costs. Even among central bankers, these large-scale asset purchases (LSAPs) and lending facilities remain somewhat controversial. A 2017 survey of central bank governors showed a plurality of respondents (38.2%) deemed that it was "too early to judge" whether QE using government debt should remain in central bank tool kits after the 2007–2009 Global Financial Crisis (GFC), while 20.6% responded that it should be discontinued (Blinder et al. 2017). An evaluation of costs poses two difficulties: (1) costs may be difficult to measure owing to the near-simultaneous announcement and implementation of multiple measures by monetary and fiscal authorities during crises, and (2) the line between intended impacts and unintended consequences from central bank facilities may be unclear if the central bank does not communicate its exact policy intentions. A growing academic literature addresses the unintended consequences and side effects of central bank crisis actions. In our paper, we review this literature.

We focus on two major forms of central bank actions: large-scale asset purchases and large-scale lending facilities.[5] Other central bank activities, such as conventional monetary policy actions, forward guidance, or regulatory actions, are outside the scope of this review. We focus on the direct costs and side effects of these facilities to the financial system, rather than addressing all possible outcomes; for example, we do not address broader macroeconomic costs or long-term moral hazard issues.

In our literature review, we highlight papers that propose mechanisms by which central bank actions may create costs or side effects as well as papers that document the existence

[5] We also briefly discuss other actions that may be undertaken as a consequence of asset purchases and lending, such as changes to reserve policies.

and magnitudes of those same effects.[6] We categorize financial system costs into five groups, which the existing literature has provided evidence in favor of:

- Impacts on market and funding liquidity (Section II),
- Pricing distortions (Section III),
- Rent-seeking and other inefficient uses of central bank liquidity (Section IV),
- International spillovers (Section V), and
- Consolidated government balance sheet distortions (Section VI).

Finally, we discuss actions central banks have taken to mitigate the costs and side effects of their intervention programs (Section VII). Figure 1 in the Appendix provides a summary of papers discussed in each section.

## II. Impacts on Market and Funding Liquidity

Central bank interventions, whether in terms of asset purchases or lending programs, are often intended to improve the liquidity of eligible assets during a crisis. A central bank provides an outlet, outside private markets, for participants to either sell or borrow against their assets. By acting as a large and committed buyer, the central bank could increase liquidity provision by dealers, reduce their inventory risk, stimulate trading activity through portfolio rebalancing, and reduce contagion effects from disorderly asset sales (Bessembinder et al. 2016; Boneva et al. 2022; Diamond and Rajan 2005). However, recent evidence suggests that the impact of these actions may not be unidirectional. For example, Boneva, Islami, and Schlepper (2024) differentiates between the short-term flow effects and long-term stock effects of central bank fixed-income purchases on market liquidity, suggesting that while interventions improve liquidity measures contemporaneously, long-term impacts may be quite different. As a central bank removes assets from the market during large-scale purchases, liquidity measures may deteriorate because fewer bonds are available in private markets. An alternative mechanism, proposed by Boermans and Keshkov (2018), suggests that central bank asset purchases tend to attract price-elastic investors who in turn sell their bonds to the central bank. This exit of price-elastic investors then leads to an increase in the concentration of bond holdings among price-inelastic investors, hampering future liquidity by reducing the quantity of bonds available for trade. In a similar vein, Ferdinandusse, Freier, and Ristiniemi (2020) shows that asset purchases are effective at initially improving liquidity, but as scarcity is induced and yields fall sufficiently to discourage new buying, liquidity begins to fall.

[6] For additional information, a more focused, policy-oriented review of these costs can be found in Logan and Bindseil (2019), while a high-level discussion of these costs is presented in Bernanke (2022).

The work above suggests that market liquidity can be significantly affected by the underlying supply (float) of bonds available for market participants. A straightforward signal that fewer bonds are available for trade following a central bank intervention is a decrease in either realized trading volumes or potential trading volumes through measures like market depth. Kurosaki et al. (2015) shows that, following the Bank of Japan's (BoJ's) quantitative and qualitative easing in the fall of 2014, the median volume of limit orders at the best-ask and best-bid price in the Japanese government bond futures market, a proxy for market depth, declined to 25% of its peak in mid-2014. Similarly, Schlepper et al. (2020) shows that order book depth in interdealer markets fell by EUR 1.3 million following a EUR 100 million German government bond purchase by the European Central Bank (ECB) through its public sector purchase programme (PSPP).

Central bank asset purchases have gone beyond government bonds and, in turn, there may be additional costs from buying corporate bonds, mortgage-backed securities (MBS) or equities. Kandrac (2018) shows that an increase in the Federal Reserve's (Fed's) stock of MBS resulted in a deterioration of trade sizes and trading volumes and the total number of trades. Specifically, the average daily amount of MBS purchased by the Fed during the reinvestment phase of the MBS program reduced trade sizes, on average, by 7.6% from the previous day, and the average purchase amount during the Fed's third phase of QE (QE3) reduced trade sizes by 5.7% from the previous day. The impact on trading volumes was, on average, an increase of 2.6% during the reinvestment phase and a decline of 3.4% during the QE3 phase.

Another group of liquidity measures are those related to bid-ask spreads. Pelizzon et al. (2018) suggests that the impact of the BoJ's QE program may have had a time component, with bid-ask spreads increasing as bonds became scarcer over the course of QE. Depending on the stage of the BoJ's QE program, bid-ask spreads for a given Japanese government bond increased between 0.33 basis points (bps) and 1.5 bps for a 1% increase in the share of the amount outstanding held by the BoJ. In the later stages of the BoJ's QE program, bid-ask spreads on a given bond changed by an additional –0.48 bp to 2.2 bps for a 1% increase in the share of the amount outstanding of that bond's close substitutes held by the BoJ. Similarly, Abidi and Miquel-Flores (2018) suggests the bid-ask spreads of corporate bonds started to increase three weeks after the announcement of the ECB's corporate sector purchase programme (CSPP), increasing by 4.4 bps in the third week and up to 17 bps by the sixth week.

Another possibility is that asset purchase programs may increase bid-ask spreads only under specific circumstances, explaining why some studies, such as Kandrac (2018) and Schlepper et al. (2020), find mixed results. It is difficult for theory to provide a satisfactory explanation for why specific bond purchase thresholds by a central bank may create negative impacts. As a result, the literature has relied on ex post analysis of different thresholds to explain the differential impacts of government purchases. Blix Grimaldi, Crosta, and Zhang (2021) finds that, in the case of the Sveriges Riksbank, negative impacts on asset volatility were up to four times higher once it had purchased at least 40% of a particular bond's outstanding quantity. Han and Seneviratne (2018) shows evidence that 40% of a bond's value outstanding being held by the BoJ corresponded to an increase in that bond's estimated bid-ask spread just above one standard deviation. However, the 40% threshold does not show universal support, as Kandrac and Schlusche (2013) finds no significant deterioration in liquidity from the flow of purchases from the Fed above this threshold. In broader terms, Boneva, Islami, and Schlepper (2024) shows that relative impacts of the ECB's German corporate bond purchases on bid-ask spreads were worse for older bonds compared with more recently issued ones, suggesting that older bonds with a lower free float were more negatively impacted.

Signs of market liquidity issues may be visible in lending markets (that is, repurchase agreement, or repo, markets) rather than for outright fixed-income transactions (that is, cash markets). A common measure of a bond's availability in the repo market is its "specialness," the difference between the general collateral repo rate and the repo rate for transactions using that specific bond. The logic of this measure is simple: if a bond is in scarce supply, borrowers who pledge this bond as collateral can borrow cash at lower rates. D'Amico, Fan, and Kitsul (2018) shows that QE purchases by the Fed increased specialness of Treasuries and that this specialness lasts up to three months. The authors find that specialness increases not only for recently issued Treasuries but also for seasoned Treasuries. Specialness would increase by 1.84 bps for the average purchase amount of recently issued Treasuries by the Fed on the day of an operation and by 0.12 bp for the average purchase size of seasoned Treasuries. The impacts of QE on repo specialness translated to lower yields on Treasuries relative to a fitted yield curve, demonstrating visible scarcity effects in both repo and cash markets for Treasuries. Arrata et al. (2020) shows that the ECB's asset purchases increased specialness by 0.78 bps for a purchase of 1% of the underlying bond's amount outstanding, while Corradin and Maddaloni (2020) shows that this specialness was also associated with a 0.32% higher probability of failures to deliver. In a similar vein, Ballensiefen, Ranaldo, and Winterberg (2023) argues that ECB asset purchases created separation in repo markets between funding-driven and collateral-driven repos, where the sensitivity of interest rates on repo transactions involving QE-eligible bonds to the ECB's benchmark repo rate was 17 bps lower than for those involving non-eligible bonds.

A known issue with lending facilities is that those who access the facilities, often bank-owned dealers, are not always the ultimate demander of liquidity. These dealers are often intermediating liquidity through matched repo transactions to other market participants such as nonbank financial institutions (NBFIs). Because these other participants typically do not have direct access to central bank facilities, dealers retain some market power in the transaction. Hüser, Lepore, and Veraart (2024) shows that although aggregate repo spreads increased during the COVID-19 crisis, these increases differed sharply across NBFI sectors, which indicates that rates from emergency facilities are not fully passing on from dealers to NBFIs. In a similar vein, the excess reserves created by crisis asset purchases may reduce incentives to participate in interbank markets and reduce policy passthrough. Akram and Findreng (2021) shows that large reserve balances after QE discourage trading and may create a positive spread in interbank markets, compared to the policy rate.

An alternative interpretation is that dealers may be unable to pass through lending as a result of their own constraints. Exploiting a regulatory change in reporting requirements of the leverage ratio that affected only a subset of dealers, Kotidis and van Horen (2018) compares repo activity between UK dealers affected by the regulatory change and thus more bound by the leverage ratio with the repo activity of unaffected UK dealers. The authors show that for a given client, repo volumes on average decreased by 66 pp, and the number of repo transactions declined by 39 pp among affected UK dealers relative to unaffected dealers.

## III. Pricing Distortions

A second category of consequences from central bank interventions is changes in the pricing of securities. Often, central bank interventions are intended, at least to some degree, to modify pricing. During a financial crisis, this may be because market participants are engaging in fire-sale behavior, artificially depressing the prices of otherwise safe assets below their fundamental values. In this case, interventions may be intended to return securities to their "fair" value. However, not all price changes caused by interventions are either intended or desirable. For example, interventions may distort long-term relationships between securities by lowering the yields of relatively riskier securities below those of safer counterparts.

Pasquariello (2018) argues that interventions may induce violations of the law of one price by complicating signals about the fundamental value of assets targeted by central banks. Similarly, Steeley (2015) argues that while the Bank of England's (BoE's) QE program narrowed bid-ask spreads, it did so at the cost of creating return anomalies, thus reducing the efficiency of market prices. In particular, between the BoE's first and third phases of QE, gilt prices were slower to adjust to the arrival of new information, as shown by higher correlations between past and present gilt returns relative to their pre-QE levels. This autocorrelation could be exploited to generate excess returns (after accounting for transaction costs) during the period between the first and second phases of QE. Barbon and Gianinazzi (2019) argues that, in Japanese equity markets, the use of price-weighted instead of value-weighted baskets by the BoJ tilted its holdings in a manner that distorted the composition of aggregate systematic risk held by the private sector. This act distorted prices by generating abnormal returns around announcements for those stocks that experienced the greatest reduction in systematic risk as a result of the BoJ's purchases.

Price distortions may be seen through changes in the prices between otherwise similar assets and modifications of arbitrage relationships. Pelizzon, Subrahmanyam, and Tomio (2025) shows that bond scarcity induced by ECB QE purchases increased transaction costs (bid-ask spreads) and the costs of borrowing securities (specialness), thereby distorting the arbitrage mechanism between futures and cash markets. Specifically, a 10% increase in the ECB's holdings of either German or Italian bonds widened the bid-ask spread of those bonds by 25% and increased their specialness to a level five times higher than it was before QE. This, in turn, corresponded to an increase of the cash-futures basis by 46 bps, which started from a level of near zero before the commencement of the ECB's QE program.[7] Lucca and Wright (2023) shows that, after the Reserve Bank of Australia's (RBA's) yield targets under

[7] The cash-futures basis is the spread between an asset's current price in the cash market and the price of the respective futures product.

its yield curve control (YCC) program were no longer consistent with expectations of the monetary policy rate path, yields of targeted bonds decoupled from the yields of other government bonds and other comparable fixed-income markets. This created kinks in the government bond yield curve because targeted bonds remained relatively lower than yields of neighboring bonds with similar maturities, which rose via the expectations channel.

Finally, another way in which a central bank may distort prices is by changing perceptions of market risk. Indeed, a central bank may wish to calm markets during a crisis by removing the possibility of tail-risk scenarios. Since assets are priced, in part, based on their potential risks, changes to these risks should result in similar changes to their prices. All else being equal, a reduction in a given asset's risk would generally be accompanied by an increase in that asset's value. Removal of the risk by the central bank can be controversial, since these actions can be seen as encouraging moral hazard on the part of market participants, who may take excessive risk when they no longer internalize tail-risk scenarios.

There is ample evidence in the literature that interventions distort prices through changes in various bond spreads, one measure of risk. On a theoretical level, Kurtzman and Zeke (2020) shows in a DSGE (dynamic stochastic general equilibrium) model that central bank purchases of corporate securities modify the spread between purchased and non-purchased securities through the portfolio channel of intermediaries. Similarly, Cimon and Walton (2024) shows theoretically that central bank purchases of one asset type create pricing changes in covariant assets through the dealer balance sheet channel.[8] Empirically, Gilchrist et al. (2024) and Boyarchenko, Kovner, and Shachar (2022) show that the Fed's Secondary Market Corporate Credit Facility (SMCCF) had different impacts on eligible bonds compared with their ineligible counterparts. Gilchrist et al. (2024) finds that corporate bonds' credit risk premiums for eligible bonds declined by 47 bps relative to ineligible bonds in a two-day window around the announcement of the Fed's SMCCF and then declined to 28 bps 10 days later. Boyarchenko, Kovner, and Shachar (2022) finds that over a three-day window around the Fed's announcement of the SMCCF, duration-matched credit spreads on A to AAA eligible bonds and BBB eligible bonds declined by 14.89 bps and 17.68 bps per day, respectively, relative to spreads of ineligible bonds. Xu and Pennacchi's (2023) analysis of the SMCCF finds a similar change in credit spreads, arguing that the exclusion criteria increased stigma around ineligible bonds. Boneva, de Roure, and Morley (2018) also finds that the BoE's corporate purchase program reduced credit spreads of an issuer's eligible bonds by 13–14 bps compared with the issuer's ineligible foreign bonds.

Using options-implied tail risk, Haddad, Moreira, and Muir (2025) shows that central bank programs reduce left-tail risk not only during but also after interventions, implying that participants internalize the possibility of future actions. Kelly, Lustig, and Van Nieuwerburgh (2016) finds a similar reduction in tail risk associated with combinations of central bank programs and government stimulus measures during 2007–2009. The authors find that price deviations between out-of-the-money put options for financial sector exchange-traded funds (ETFs) and corresponding options on the underlying companies increase by 31% in

[8] The dealer balance sheet channel is the way by which dealers' balance sheet costs and frictions impact trading and prices in the markets they intermediate.

the first five days following positive announcements of both central bank interventions and fiscal stimulus for the financial sector, demonstrating that the market perceived these interventions as sector-wide tail-risk insurance.

## IV. Rent-Seeking and Other Inefficient Uses of Central Bank Liquidity

A concern regarding interventions by central banks is that they can create opportunities for market intermediaries to extract economic rents, which are ultimately borne by investors and taxpayers. Dealers who intermediate over-the-counter (OTC) markets may find themselves with a new, reliable counterparty in the form of the central bank. This reliability is intentional, as it allows dealers to respond more efficiently to liquidity demand by their clients, preventing market dysfunction; however, it can also be used as a source of rents. A related concern is that firms and other market participants who receive liquidity and funding from these dealers may use it for inefficient purposes. This latter concern can arise if firms are not fully internalizing the riskiness of their activities, due to their own expectation of central bank support.

Excess rent extraction by dealers can arise from the design of auctions and facilities through which central banks implement their interventions. One design choice that can create profit opportunities for dealers is the central bank's rules for allocating the auctioned amounts across dealers' bids. Song and Zhu (2018) suggests that since dealers could partially infer the specific bonds and prices that the Fed was willing to accept, dealers could short specific bonds into the Fed's QE auctions at a profit and later close their positions in private markets. The authors estimate that one standard deviation in the "cheapness" of a bond (the bond's yield relative to a model-fitted yield) induced the Fed to purchase $276 million more of that bond and increased the auction cost on that bond by 2.6 cents per $100 par value. Similarly, Breedon (2018) finds that round-trip transaction costs, the difference between bond yields at auction and yields on days around the auction, totaled 0.5% of the total value of the BoE's QE program, and that some of this cost was attributed to dealers' being able to exploit the allocation rules of the BoE's QE auctions.

Profit opportunities can also arise from the predictable trade sizes of the central bank and the publicly known set of dealers a central bank transacts with. An and Song (2023) estimates that, in the Fed's agency MBS auctions, dealers charged the Fed higher than private market clients by 2.5 cents per $100 par value, representing half of dealers' average gross profit margin. The authors find some evidence that this higher markup for the Fed was a result of the Fed's larger trade sizes compared with those of private market clients and 'its fixed set of dealer counterparties. In a study of the BoE's QE auctions, Boneva, Kastl, and Zikes (2025) suggests that dealers earned a rent of 2.6 bps intermediating such auctions. The authors suggest that when a central bank intends to provide liquidity to dealers in an equitable way, no mechanism could fully eliminate these rents that dealers can extract because of their informational advantages.

Apart from intermediaries, security issuers may be enticed by the low rates caused by central banks' actions to engage in inefficient uses of liquidity provided by the central bank. Todorov (2020) shows that firms that issued bonds eligible for the ECB's CSPP used these issuances to increase corporate dividends four times higher in the two quarters following the announcement of the CSPP, relative to the first quarter before the announcement. At the same time, the cash holdings, working capital, research and development, and property, plant, and equipment accounts of these firms did not change. However, given data limitations, the author cautions against interpreting these results as purely a result of the ECB's CSPP. De Santis and Zaghini (2021) analyzes the same program and shows that while firms did increase investment in capital expenditures and intangibles, eligible firms also repurchased their own securities and held more cash. Darmouni and Siani (2025) shows that corporations that were more exposed to the Fed's corporate credit facilities during the COVID-19 crisis were more likely to increase cash and less likely to increase real investments, perhaps because these firms were not initially financially constrained. Strikingly, for every dollar of new bonds issued, 45 cents were used to increase cash, 15 cents to pay back bank debt, and 30 cents to refinance existing bonds, with no increase in real investment.

Firms may also use central bank facilities to increase leverage or to issue securities with riskier profiles. This behavior can also be seen as creation of moral hazard by the central bank, as firms take advantage of an artificially low price of risk to engage in actions that would otherwise be more expensive. Indeed, Foley-Fisher, Ramcharan, and Yu (2016) shows that, in response to the Fed's maturity extension program (MEP), firms with a preference for long-term debt, as proxied by the share of total debt that matures beyond three years, chose to issue additional long-term debt. During the MEP period, a one-standard-deviation increase in the authors' proxy for long-term debt preference corresponded to an 8 pp increase in long-term debt growth. Pegoraro and Montagna (2021) shows that firms took advantage of decreased risk premiums and increased issuance of riskier bonds in response to the ECB's CSPP: more of these bonds were unsecured or non-guaranteed, with longer maturities, more fixed coupons, and a more general (rather than specific) purpose. Similarly, Kandrac and Schlusche (2021) shows that excess central bank reserves from QE do result in an increase in bank lending, but that these banks increase their share of riskier loans. Bond characteristics aside, Acharya et al. (2025) shows that in response to Fed QE programs during the GFC, firms vulnerable to a credit rating downgrade increased their bond issuance and used the proceeds to engage in more merger and acquisition activity. The authors show that this was riskier and prolonged rating downgrades that occurred during the COVID-19 crisis.

These impacts may be felt outside the country engaging in QE, suggesting that central banks may create morally hazardous scenarios outside their jurisdictions. Morais et al. (2019) shows that quantitative easing in the United States and United Kingdom created spillovers in Mexican markets, this time through the bank-lending channel.[9] In response to a one-standard-deviation increase in the Fed's assets relative to GDP, a proxy for QE, US banks

[9] The bank-lending channel is the method by which banks' lending choices impact financial markets and financial market participants.

operating in Mexico experienced 6.5% higher future loan defaults. Moreover, one standard deviation in the proxy for QE led to an 8.6% increase in future loan defaults for firms with loans that had higher than average ex ante interest rates––evidence consistent with the interpretation that QE induced banks to engage in reach-for-yield behavior by lending to riskier firms.

## V. International Spillovers

With mandates originating in their home countries, central banks from large economies trade off restoring functioning in their financial markets against international spillovers from their policies. These spillovers have been shown to be meaningful across both advanced and emerging market economies (EMEs). When financial stress is global, financially easing spillovers from one central bank's extraordinary actions may spare other central banks from taking similar actions. However, when the economic stance across countries differs, spillovers may undermine central banks' ability to tailor policies to their home economies.

Theoretical work by Gourinchas, Ray, and Vayanos (2025) and Greenwood et al. (2023) rationalizes the response of exchange rates and foreign bond yields through the risk-bearing capacity of global investors. In these models, foreign currency and bond risk premiums decline because global investors are willing to bear more risk as they hold less supply of home country bonds following central bank purchases. A clear example of this mechanism comes from the GFC period: Neely (2015) shows that major country sovereign bond yields changed by an average of 42% of the change in US Treasury yields across the Fed's LSAP announcements over 2008–2009. At the same time, the cumulative depreciation of the US dollar against major currencies was –10.15% across eight news events about the Fed's LSAP program from 2008–2009. Apart from the Fed's actions, Joyce, Tong, and Woods (2011) finds that the Sterling Exchange Rate Index—the trade-weighted value of the sterling—declined by 4% around six of the BoE's QE-related announcements over 2009–2010. Georgiadis and Gräb (2016) and Fukuda (2017) attribute similar exchange rate depreciations to extraordinary asset purchase announcements by the ECB and BoJ.

The degree of integration between EMEs and advanced economies may impact the magnitude of spillovers. MacDonald (2017) finds that EMEs whose capital markets were more highly integrated with those of the US saw larger currency and equity price appreciations over 2008–2014, when the Fed bought US Treasury securities as part of its LSAP program. For a 1% rise in LSAPs, the appreciation of the currency of a country with close economic ties to the US, such as Mexico, was double that of a country with low ties to the US, such as Indonesia. Placing the size of these movements in context, Bowman, Londono, and Sapriza (2015) argues that the spillovers from the Fed's unconventional policies to EME sovereign bond yields are not very different than the usual response of bond yields to US monetary policy shocks, when controlling for an EME's' exposure to US financial conditions.

EMEs are also the recipient of global capital inflows following unconventional monetary policy easing in advanced economies. Fratzscher, Lo Duca, and Straub (2018) finds EME countries experienced equity inflows of 10.8% and bond inflows of 21%, while other advanced economies saw smaller inflows of 5.4% and 12.6% in their equity and bond markets from the Fed's QE2 and QE3. EMEs received larger capital inflows following the Fed's QE announcements when US macroeconomic uncertainty was lower and the US economic outlook was positive. Lo Duca, Nicoletti, and Martínez (2016) explains that such capital inflows led to higher bond issuance by nonfinancial corporates in both advanced economies and EMEs. The authors estimate that without the Fed's QE over the GFC, nonfinancial corporate issuance in EMEs would have been half the amount issued over 2009–2013.

Capital flows may constrain EMEs in their ability to conduct monetary policy. In the model of Anaya, Hachula, and Offermanns (2017), EMEs limited spillovers from the US by lowering interest rates to align their monetary policies with the easing stance of the US. Consistent with Rey (2018), policies by the financial centers of the world have an outsized impact on capital flows to smaller countries, which limits their ability to conduct monetary policy.

Reversing unconventional monetary policy though quantitative tightening (QT) has also been shown to have larger impacts on capital flows than policy easing. During the initial phases of Fed QT from 2013–2014, Chari, Stedman, and Lundblad (2021) estimates, bond and equity flows into EMEs were twice as sensitive to monetary policy shocks than they were over the Fed's QE period.

## VI. Consolidated Government Balance Sheet Distortions

The final class of issues we address relates to the consolidated government balance sheet; that is, the combined balance sheet of the fiscal authority and the central bank. A broad discussion of these issues comes from Rogoff (2021), which outlines threats to the central bank's independence and credibility as a result of asset purchases. The need for central banks to purchase large volumes of government debt may subsidize fiscal stimulus and complicate roles between the monetary and fiscal authorities. We discuss these issues from two standpoints: the impact of central bank asset purchases on the fiscal authority and the impact of potential losses from asset purchases on the central bank.

Central bank actions may have a direct impact on the incentives of government treasury managers. Hodula and Melecký (2020) argues that central bank crisis actions may incentivize government debt managers to lower the maturity profile of their debt. Ultimately, this shorter maturity debt may increase their susceptibility to rollover risk and increases in funding costs if rates rise.

In a general sense, government asset purchases by the central bank create a distortion between the duration of the debt issued by the fiscal authority and the duration of the consolidated debt of the government, including the central bank. This is because central banks often purchase long-dated government bonds and issue floating-rate reserves, shortening the duration profile of the consolidated government balance sheet. Greenwood et al. (2014) argues that this distortion may push against government attempts to lengthen their maturity profiles during a crisis. To that end, Fortin (2022) shows in Canada the published average debt maturity of the fiscal authority decreased from 75 months to 58 months during the COVID-19 crisis, before recovering to 78 months in 2022. When considering the Bank of Canada's (BoC's) liabilities, the consolidated debt maturity decreased to 40 months in 2021, recovering to only 57 months in 2022. Hooley et al. (2023) argues that this decrease in consolidated debt duration increases risks to the central bank's profitability if rates rise after a crisis, a topic to which we now turn.

A central bank that purchases bonds during a crisis may face losses if it raises rates afterward. If its purchases are large in scale, these losses can ultimately be very large in magnitude. Hall and Reis (2015) argues that a theoretical insolvency risk exists if fiscal authorities are unwilling to recapitalize an unprofitable central bank. As a result, Bhattarai, Eggertsson, and Gafarov (2015) and Hooley et al. (2023) argue that a central bank that is averse to losses may have an incentive to keep rates low following a crisis.

There is strong support that central bank losses may primarily be viewed through the lens of political economy. Bell et al. (2023) highlights that some central banks have historically reported ongoing losses but that problems may occur if miscommunication jeopardizes the central bank's credibility. Christensen, Lopez, and Rudebusch (2015) argues that the Fed could simply recapitalize itself through seigniorage, if necessary, though this could result in political economy costs. In similar veins, Belhocine, Bhatia, and Frie (2023), Goncharov, Ioannidou, and Schmalz (2023) and Adrian et al. (2024) all cite political economy issues as risks to a central bank that incurs losses.

## VII. Mitigation of Costs and Side Effects

The existence of costs from central bank actions does not imply that central banks are unable to mitigate or manage them. On the contrary, central banks can take several actions to reduce these side effects. In a report prepared for the Bank for International Settlements Markets Committee, Logan and Bindseil (2019) categorizes these mitigants into three groups:

- Program design (purchase protocols, lending program protocols, transparency, minimizing uncertainty, minimizing information asymmetry, and flexibility of implementation),
- Securities-lending programs (primary, secondary, and backstop), and
- Liability management practices and remuneration policies (absorbing excess liquidity, floor systems, and reserve remuneration tiering).

A key element of program design is the choice and quantity of securities purchased by the central bank. When securities are purchased according to a specific index composition (for example, price-weighted versus value-weighted), private markets may be distorted (Barbon and Gianinazzi 2019). Duffie and Keane (2023) argues that central banks can reduce distortion through the use of "delivery choice" reverse auction designs, which share some characteristics with "product mix auctions" (Klemperer 2010). In this design, central banks do not make offers on specific securities; instead, bidders are able to offer any eligible security at a spread to some maturity benchmark. Such a system can reduce inefficiencies from misallocated securities where scarcities exist among participants. Duffie and Keane also argues in favor of transparent programs, which limit political economy costs and rent-seeking by eligible participants.

As a more specific measure, central banks may be able to mitigate collateral shortages and mispricing of specific bonds by engaging in securities-lending programs. Broadly speaking, these programs lend out securities on the central bank's balance sheet to market participants, increasing the supply of these securities in the private market. Use of these programs by the ECB has been shown to reduce bond scarcity that results from QE programs and increase the supply of collateral available in private markets (Greppmair and Jank 2023; Pelizzon, Subrahmanyam, and Tomio 2025). Similar programs in Canada have been shown to mitigate the negative consequences of settlement failures (Fontaine, Garriott, and Gray 2016).

Some central banks have turned to tiered reserve remuneration to limit the excess quantity of reserves created by asset purchases. Altavilla et al. (2022) and Fuster, Schelling, and Towbin (2024) show that reserve tiering increases lending from those with excess reserves to those with reserves below tiering thresholds. Basten and Mariathasan (2023) and Boutros and Witmer (2020) show that reserve tiering can increase the passthrough central bank policy rates, in empirical and theoretical settings, respectively. Combining the two effects, Akram and Findreng (2021) shows that tiered reserves may both increase trading in the interbank market and decrease spreads between the interbank market and the target rate.

In some circumstances, measures taken by central banks or financial market infrastructure providers treat the symptoms of these costs, rather than addressing the underlying causes. One example of such a measure is a fail fee for repo transactions, where parties are charged a daily fee if they fail to deliver the security. If the costs of repo failures are greater on the broader financial system than they are to individual participants, fail fees may be seen as mitigating moral hazard on the part of market participants by increasing the individual cost associated with failing to deliver. Indeed, the introduction or increase of a fail fee has been shown to reduce the probability of failures to deliver (Corradin and Maddaloni 2020). However, such a fee simply discourages parties from failing to deliver bonds as a result of scarcity but does not resolve the bond scarcity itself.

## VIII. Conclusion

It is clear from existing literature that, despite the benefits, there are unintended costs, distortions, and side effects of central bank interventions. While it is possible that the benefits outweigh the costs, based on the extreme economic costs of financial crises, it is important for policymakers to be cognizant of the large unintended consequences they may provoke. This paper set out to categorize these potential unintended costs and to review recent literature that provides evidence of their existence. The full economic impacts of these interventions are still not understood, and there is ample space for future work to that end.

Future work could analyze whether the side effects and unintended costs of central bank actions are attenuated or in fact exacerbated when they interact with NBFIs rather than dealers. Traditionally, central banks have often intervened through a network of bank-owned dealers. These dealers have played a central role in fixed-income markets and are often involved with central banks through mechanisms such as the issuance of government debt. Since these dealers are often subsidiaries of commercial banks, they are also highly regulated entities and may be viewed as relatively safe counterparties. However, the ultimate target of central bank actions may not be these dealers; instead, central banks may satisfy liquidity demand by NBFIs such as pension funds, insurance companies, mutual funds, or hedge funds. Indeed, d'Avernas, Vandeweyer, and Darracq Pariès (2020) argues that the existence of a large NBFI sector may limit the reach of traditional central bank programs. Some central banks have facilities that cater directly to these participants. For example, in 2020, the BoC introduced the Contingent Term Repo Facility to lend to nonbank counterparties; while in 2023, the BoE announced plans for lending tools for insurance companies and pension funds (Hauser 2023), which culminated in the 2024 introduction of the Contingent Non-Bank Financial Institution Repo Facility.

Another area of interest is the impact of changes to market structure on the costs of central bank actions. Traditionally, fixed-income securities were often traded exclusively through dealers with little transparency, even post-trade. If the securities targeted by central banks move toward an all-to-all market structure, are centrally cleared, or are exchange traded, the balance of costs from these interventions could change. Alternatively, the size and scope of central bank interventions may be fundamentally different for primarily exchange-traded securities or securities that are intermediated by nonbank dealers.

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# X. Appendix

## Figure 1: Summary of Papers, by Topic Area

| Subtopic | Study | Central Bank(s) Studied | Facility Type(s) |
|---|---|---|---|
| **Market Liquidity** | | | |
| Trading Volumes and Market Depth | Boermans and Keshkov (2018) | ECB | GBP |
| | Boneva, Islami, and Schlepper (2024) | ECB | NBP |
| | Ferdinandusse, Freier, and Ristiniemi (2020) | ECB, Th | GBP |
| | Kandrac (2018) | Fed | MBS |
| | Kurosaki et al. (2015) | BoJ | GBP |
| | Schlepper et al. (2020) | ECB | GBP |
| Transaction Costs (Bid-Ask Spreads) | Abidi and Miquel-Flores (2018) | ECB | NBP |
| | Blix Grimaldi, Crosta, and Zhang (2021) | SR | GBP |
| | Boneva, Islami, and Schlepper (2024) | ECB | NBP |
| | Han and Seneviratne (2018) | BoJ | GBP |
| | Kandrac (2018) | Fed | MBS |
| | Kandrac and Schlusche (2013) | Fed | GBP, MBS |
| | Pelizzon et al. (2018) | BoJ | GBP |
| | Schlepper et al. (2020) | ECB | GBP |
| Repo Market Specialness | Arrata et al. (2020) | ECB | GBP |
| | Ballensiefen, Ranaldo, and Winterberg (2023) | ECB | GBP |
| | Corradin and Maddaloni (2020) | ECB | GBP |
| | D'Amico, Fan, and Kitsul (2018) | Fed | GBP |
| Pass Through of Repo Facilities | Akram and Findreng (2021) | NB | RP |
| | Hüser, Lepore, and Veraart (2024) | BoE | CL |
| | Kotidis and van Horen (2018) | BoE | Gen |
| **Asset Pricing Distortions** | | | |
| Changes in Prices | Barbon and Gianinazzi (2019) | BoJ | EP |
| | Lucca and Wright (2023) | RBA | GBP |
| | Pasquariello (2018) | Th | GBP, NBP, EP |
| | Pelizzon, Subrahmanyam, and Tomio (2025) | ECB | GBP, SL |
| | Steeley (2015) | BoE | GBP |
| Changes in Risk | Boneva, de Roure, and Morley (2018) | BoE | NBP |
| | Boyarchenko, Kovner, and Shachar (2022) | Fed | NBP |
| | Cimon and Walton (2024) | Th | GBP, NBP |
| | Gilchrist et al. (2024) | Fed | NBP |
| | Haddad, Moreira, and Muir (2025) | Fed, Th | NBP |
| | Kelly, Lustig, and Van Nieuwerburgh (2016) | Fed | Gen |
| | Kurtzman and Zeke (2020) | Th | GBP, NBP |
| | Xu and Pennacchi (2023) | Fed | NBP |

| | | | |
|---|---|---|---|
| **Rent-Seeking and Inefficient Uses of Interventions** | | | |
| Dealer Rent-Seeking | An and Song (2023) | Fed | MBS |
| | Boneva, Kastl, and Zikes (2025) | BoE | GBP |
| | Breedon (2018) | BoE | GBP |
| | Song and Zhu (2018) | Fed | GBP |
| Inefficient Uses of Interventions | Acharya et al. (2025) | Fed | GBP, MBS |
| | Darmouni and Siani (2025) | Fed | NBP |
| | De Santis and Zaghini (2021) | ECB | NBP |
| | Foley-Fisher, Ramcharan, and Yu (2016) | Fed | GBP |
| | Kandrac and Schlusche (2021) | Fed | Gen |
| | Morais et al. (2019) | Survey | Gen |
| | Pegoraro and Montagna (2021) | ECB | NBP |
| | Todorov (2020) | ECB | NBP |
| **International Spillovers** | | | |
| Exchange Rates and Bond Yields | Fukuda (2017) | BoJ | GBP, NBP, EP |
| | Georgiadis and Gräb(2016) | ECB | GBP |
| | Gourinchas, Ray, and Vayanos (2025) | Th | GBP |
| | Greenwood et al. (2023) | Th | GBP |
| | Joyce, Tong, and Woods (2011) | BoE | GBP, NBP |
| | Neely (2015) | Fed | GBP, MBS |
| Capital Flows | Anaya, Hachula, and Offermanns (2017) | Fed | GBP, MBS |
| | Bowman, Londono, and Sapriza (2015) | Fed | GBP, MBS |
| | Chari, Stedman, and Lundblad (2021) | Fed | GBP, MBS |
| | Fratzscher, Lo Duca, and Straub (2018) | Fed | GBP, MBS |
| | Lo Duca, Nicoletti, and Martínez (2016) | Fed | GBP, MBS |
| | MacDonald (2017) | Fed | GBP, MBS |
| **Consolidated Government Balance Sheets** | | | |
| Treasury Issuance Incentives | Fortin (2022) | BoC | GBP |
| | Greenwood et al. (2014) | Fed | GBP |
| | Hodula and Melecký (2020) | CNB | Gen |
| | Hooley et al. (2023) | Survey | Gen |
| | Rogoff (2021) | Th | Gen |
| Central Bank Losses | Adrian et al. (2024) | Th | GBP |
| | Belhocine, Bhatia, and Frie (2023) | ECB | Gen |
| | Bell et al. (2023) | Survey | Gen |
| | Bhattarai, Eggertsson, and Gafarov (2015) | Th | GBP |
| | Christensen, Lopez, and Rudebusch (2015) | Fed | GBP, MBS |
| | Goncharov, Ioannidou, and Schmalz (2023) | Survey | Gen |
| | Hall and Reis (2015) | Fed, ECB, Th | GBP, MBS, CL |
| | Hooley et al. (2023) | Survey | Gen |

| Actions to Mitigate Costs | | | |
|---|---|---|---|
| Facility Design | Barbon and Gianinazzi (2019) | BoJ | EP |
| | Duffie and Keane (2023) | Survey | GBP |
| | Klemperer (2010) | Th | Gen |
| | Logan and Bindseil (2019) | Survey | Gen |
| Securities-Lending Facilities | Fontaine, Garriott, and Gray (2016) | BoC | SL |
| | Greppmair and Jank (2023) | ECB | SL |
| | Logan and Bindseil (2019) | Survey | Gen |
| | Pelizzon, Subrahmanyam, and Tomio (2025) | ECB | GBP, SL |
| Repo Fail Fees | Corradin and Maddaloni (2020) | ECB | GBP |
| Reserve Tiering | Akram and Findreng (2021) | NB | RP |
| | Altavilla et al. (2022) | ECB | RP |
| | Basten and Mariathasan (2023) | SNB | RP |
| | Boutros and Witmer (2020) | Th | RP |
| | Fuster, Schelling, and Towbin (2024) | SNB | RP |

**Central banks studied:** Bank of Canada (BoC), Bank of England (BoE), Bank of Japan (BoJ), Czech National Bank (CNB), European Central Bank and Eurosystem (ECB), Federal Reserve (Fed), Norges Bank (NB), Reserve Bank of Australia (RBA), Swiss National Bank (SNB), Sveriges Riksbank (SR), survey of three or more central banks (Survey), theoretical or general discussion (Th).

**Facility types:** cash lending (CL), securities lending (SL), government bond purchase (GBP), nongovernment bond purchase (NBP), mortgage-backed security purchase (MBS), secondary-market equity purchase (EP), reserve policy (RP), general or nonfacility discussion (Gen).

*Source: Authors' analysis.*